CS 412 / 512
Introduction to Computer Graphics

Zhu Wang
Assistant Professor of Computer Science

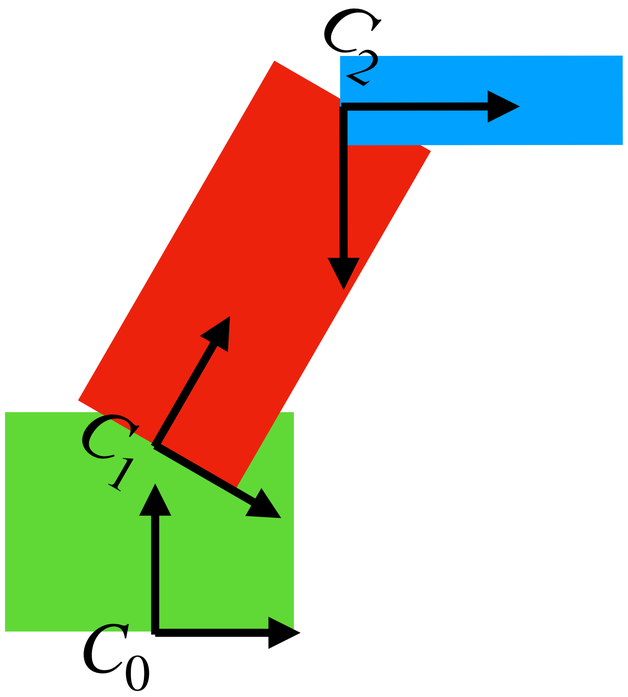


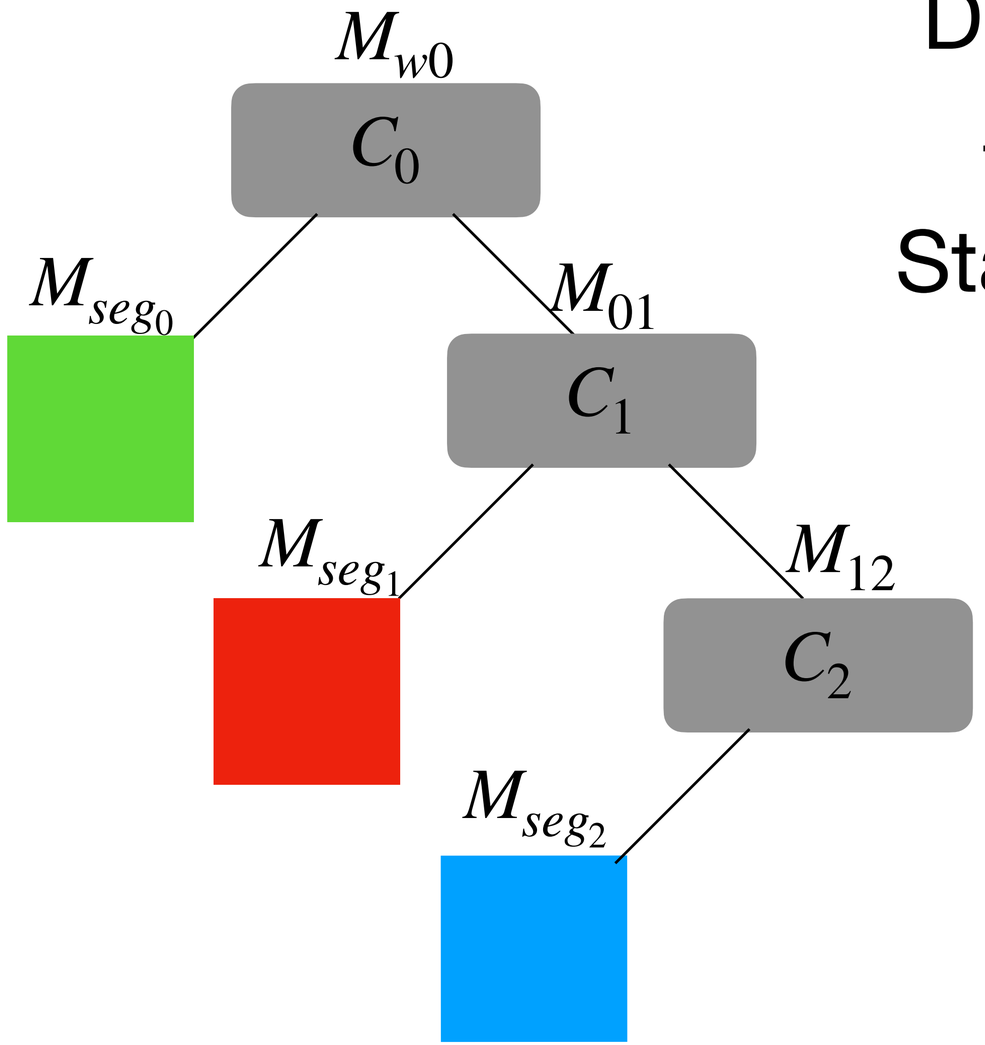
COMPUTER SCIENCE

Hierarchical Modeling

Data Structure

Tree and stack





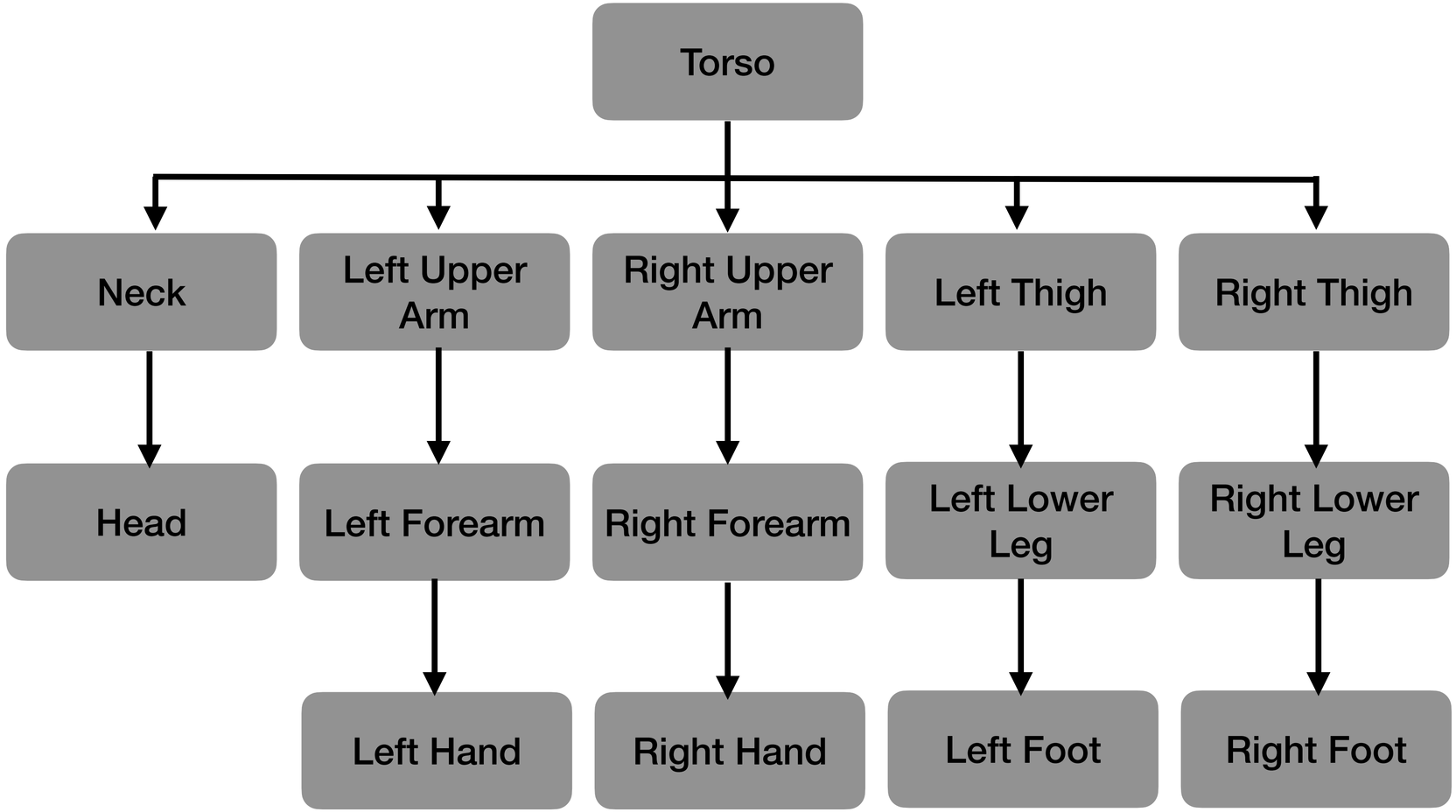
DFS

+

Stack

Hierarchical Modeling

Hierarchy of Humanoid robots



Hierarchical Modeling

Logic of the process

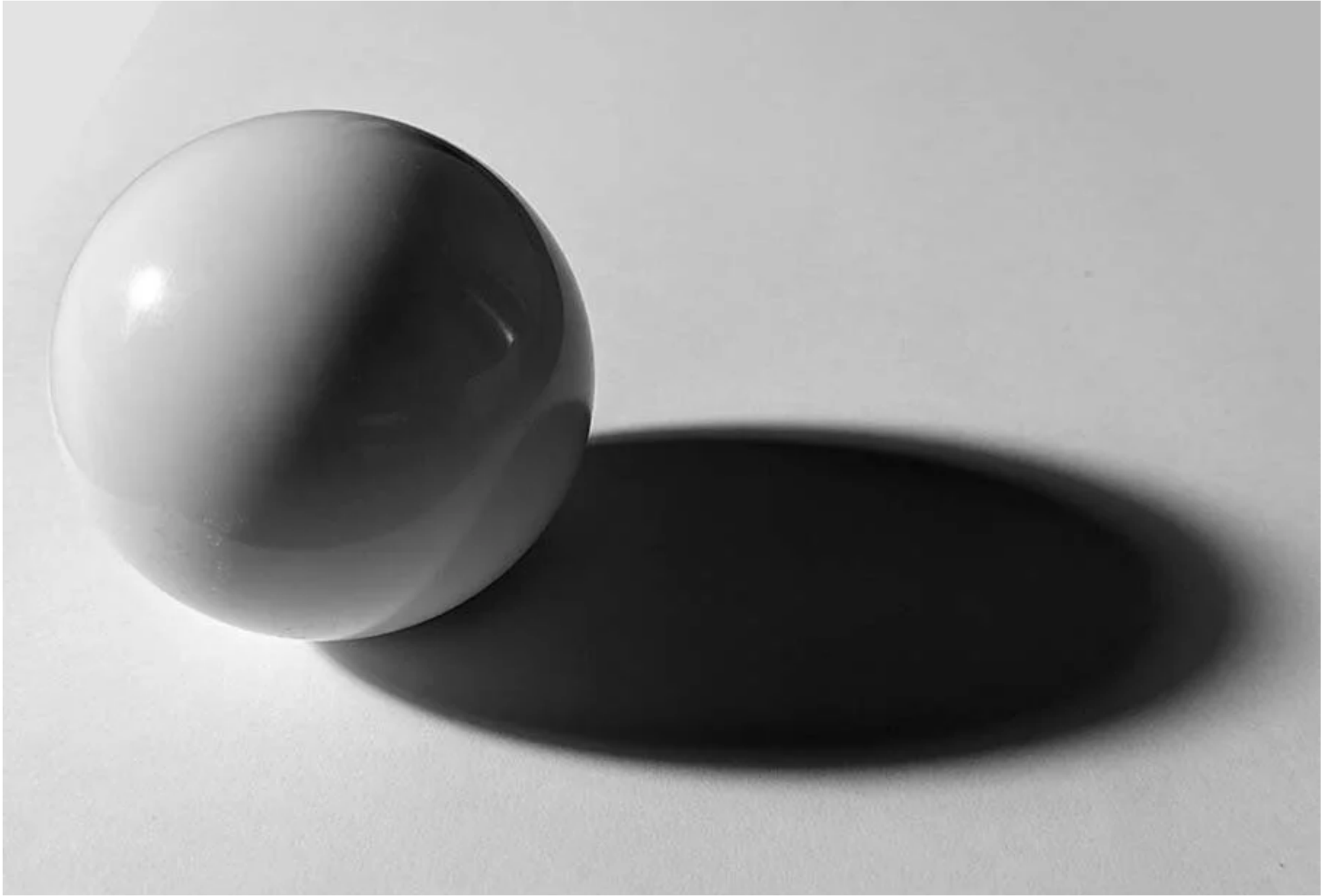
```
pushMatrix();        // world -> torso frame
        applyFrameTransform(Torso);
        drawTorso();

        pushMatrix();    // torso -> neck frame
                applyFrameTransform(Neck);
                drawNeck();

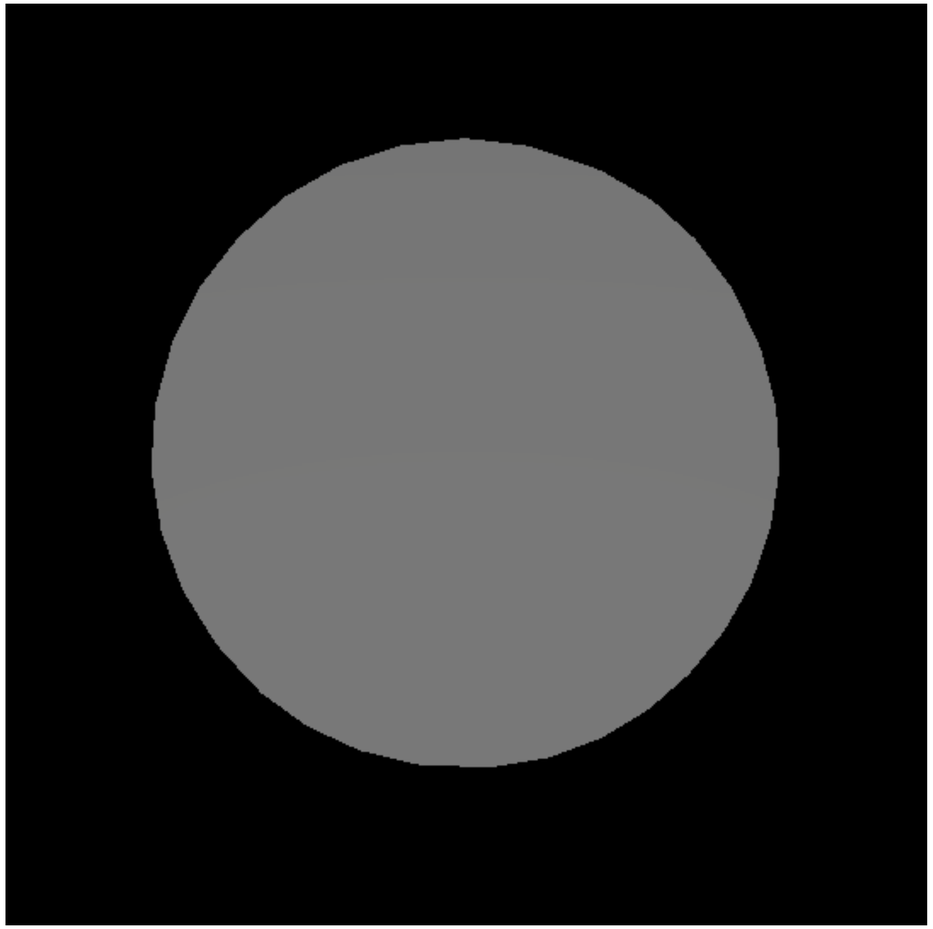
                pushMatrix();      // neck -> head frame
                        applyFrameTransform(Head);
                        drawHead();
                popMatrix();       restore neck frame
        popMatrix();    // restore torso frame

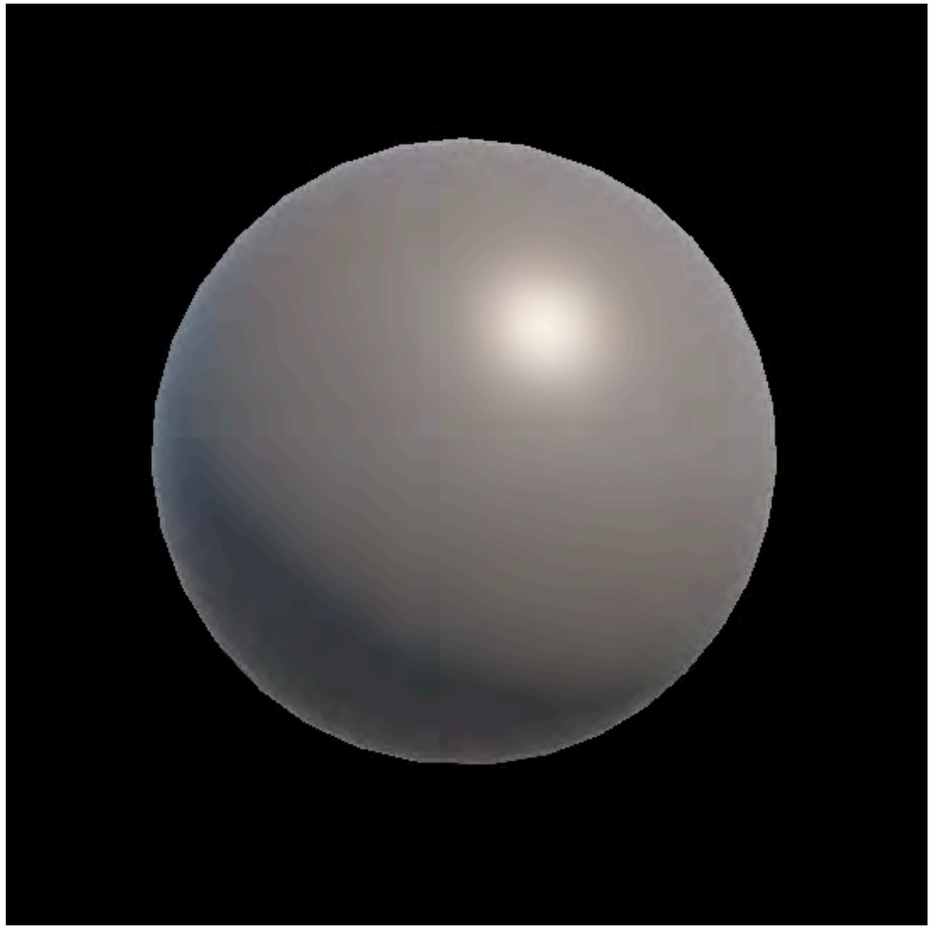
                        // same for limbs
                                ...
popMatrix();    //restore world frame
```

Lighting & Shading

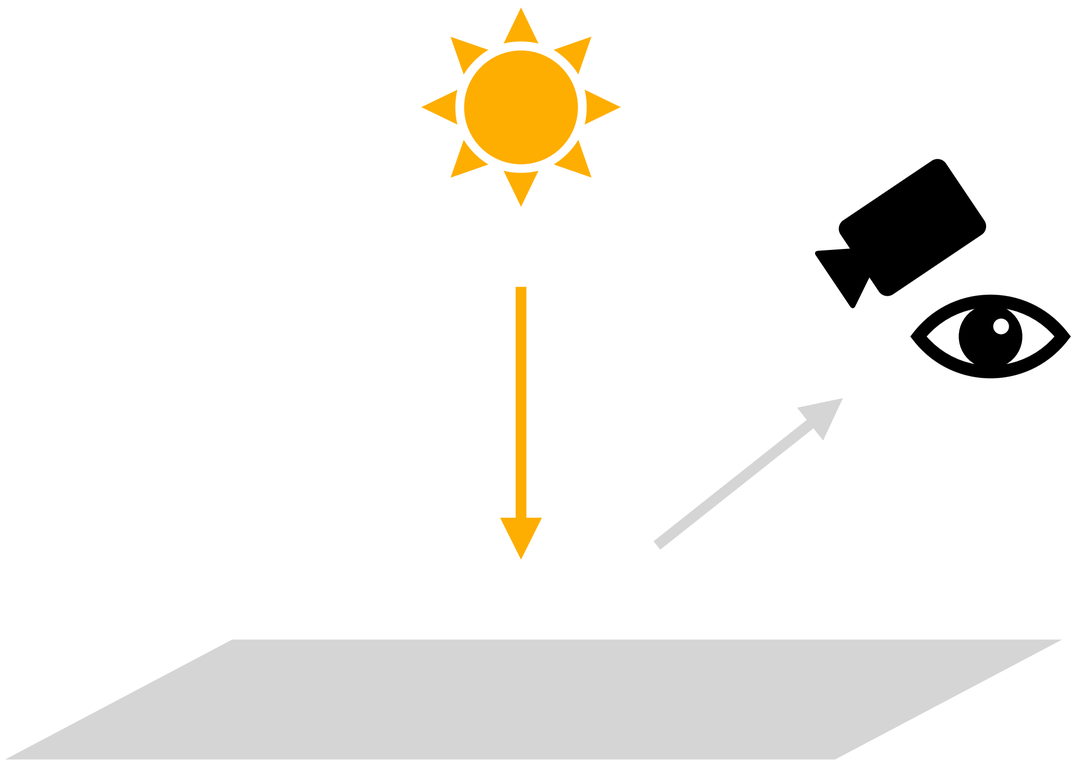


Lighting & Shading





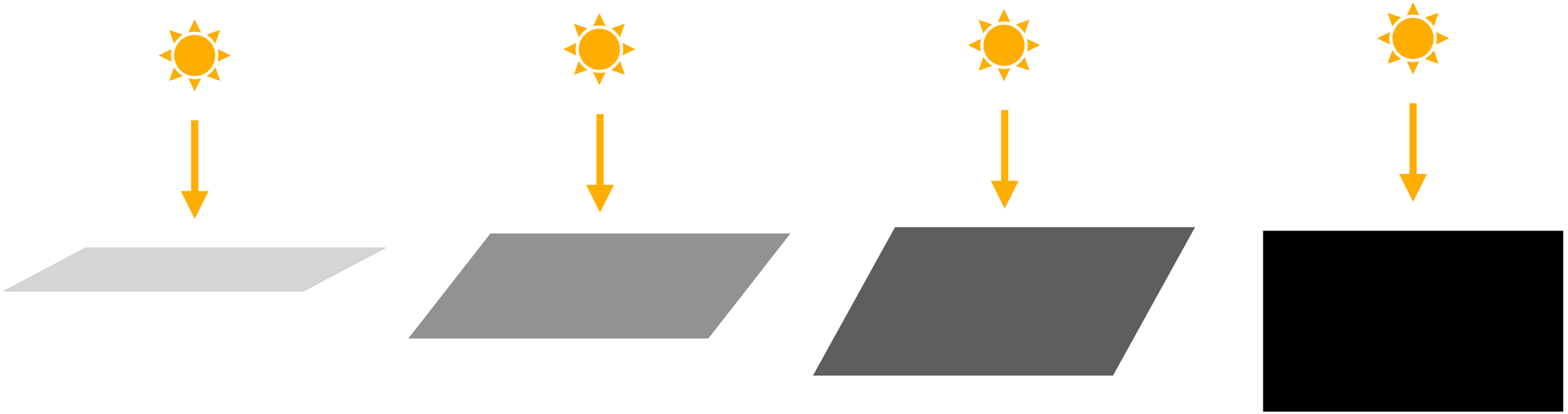
Shading



- **Light source:**
  position, direction, intensity
- **Surface properties:**
  color, reflectivity, shininess
- **Viewing direction**

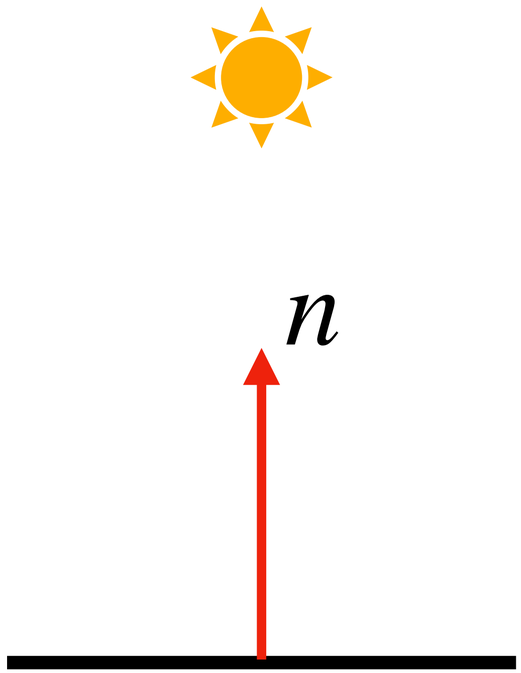
Shading

Tilt relative to the light changes appearance

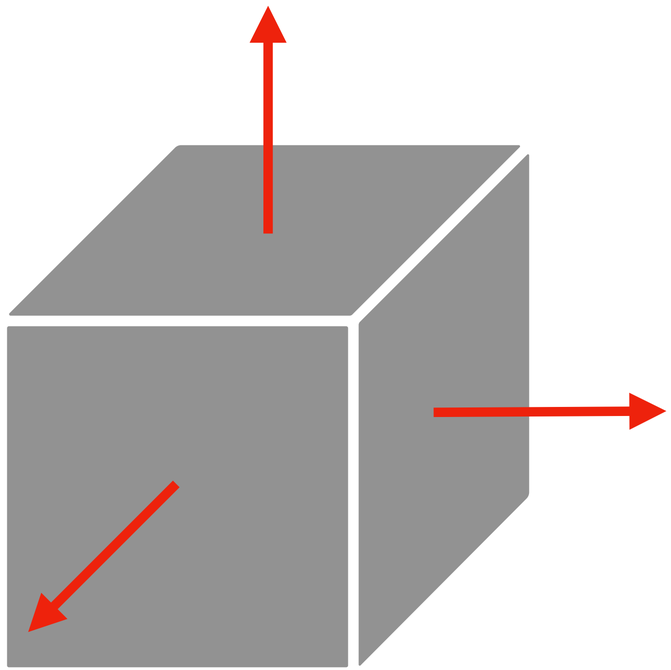


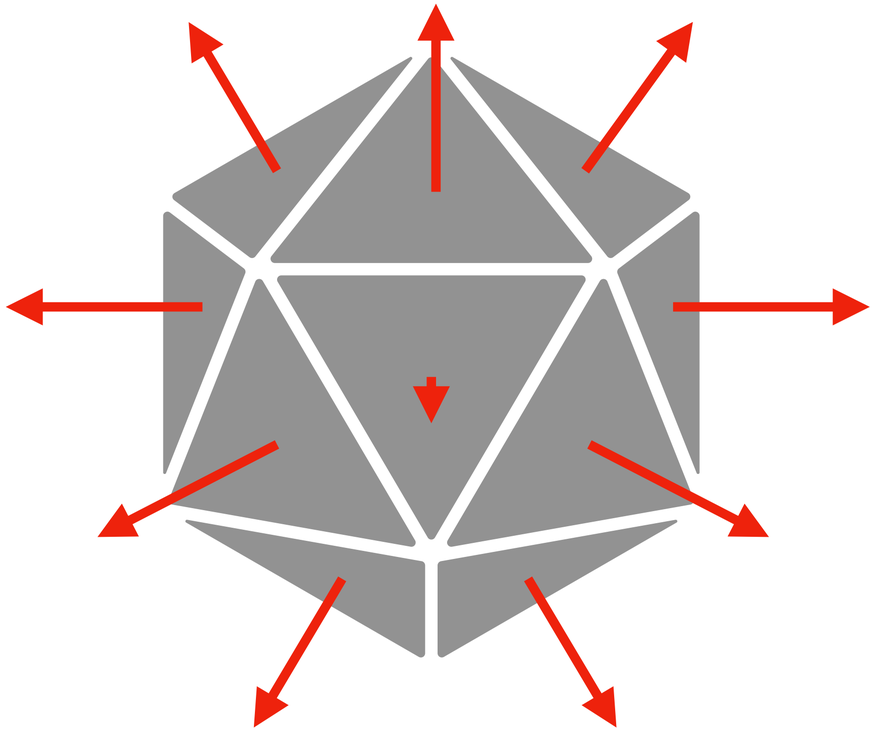
Normal vector

A normal is a vector that is perpendicular to the surface at a specific point



Normal vector

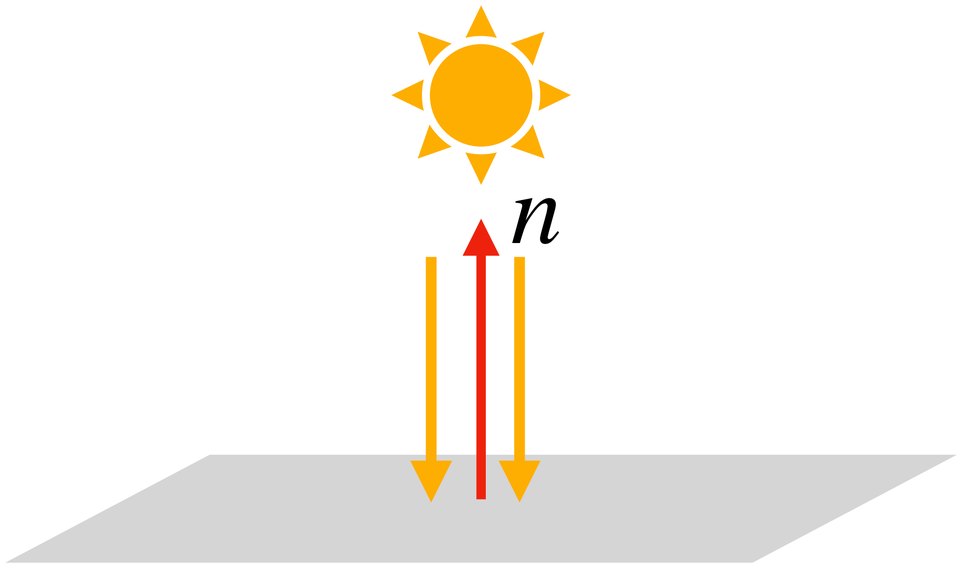


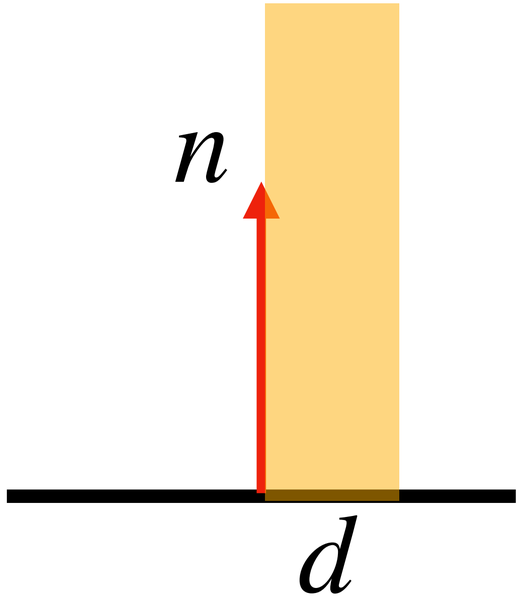


Normals are the same on the same flat surface

Different at points on a curvature

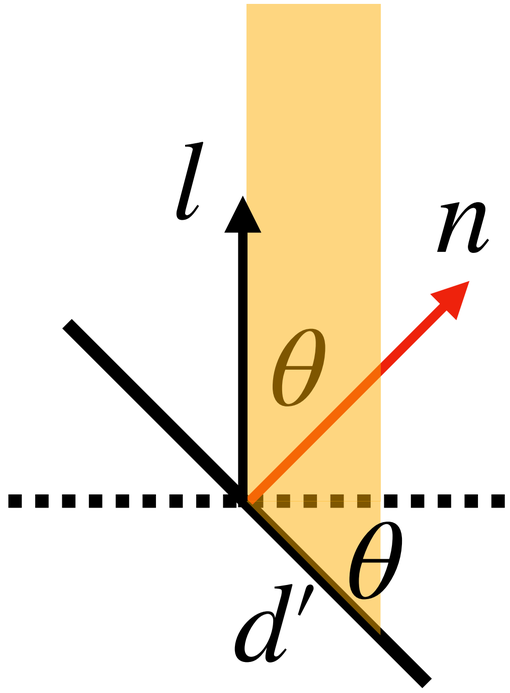
Lighting





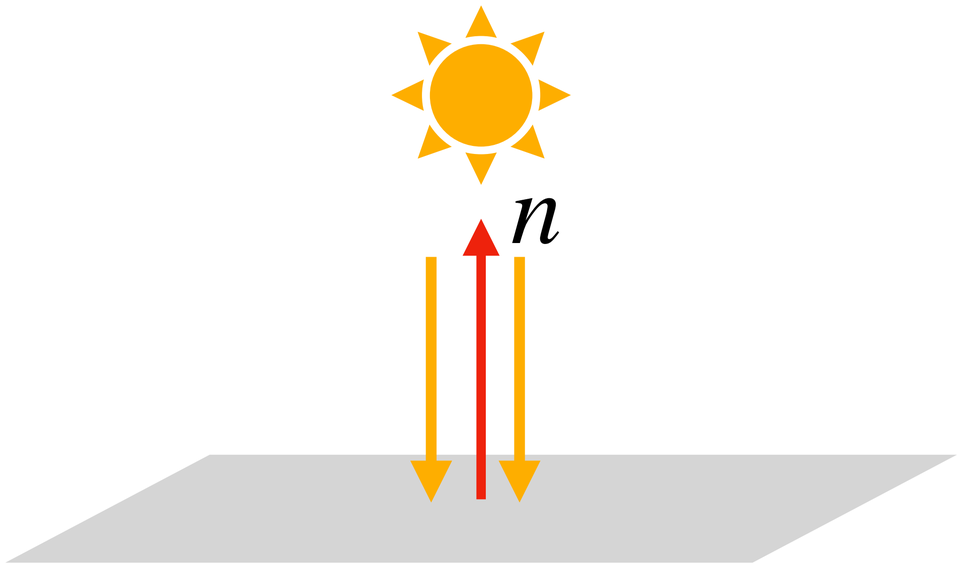
L Amount of light

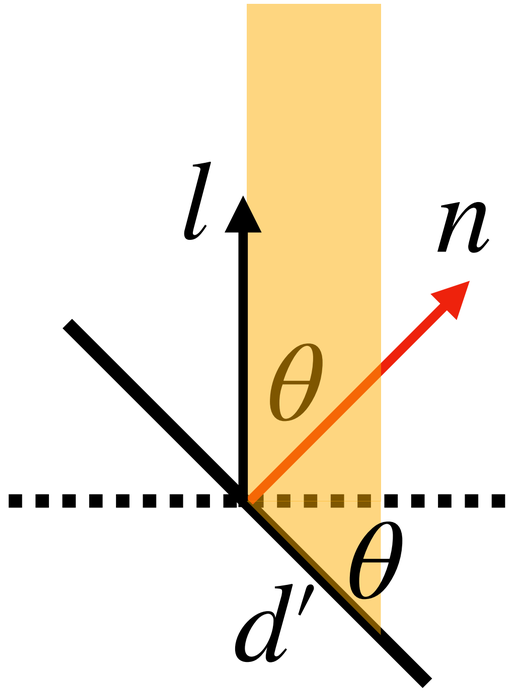
$$d = d'cos\theta$$



$$\frac{L}{d'} = \frac{L}{d}cos\theta$$

Lighting



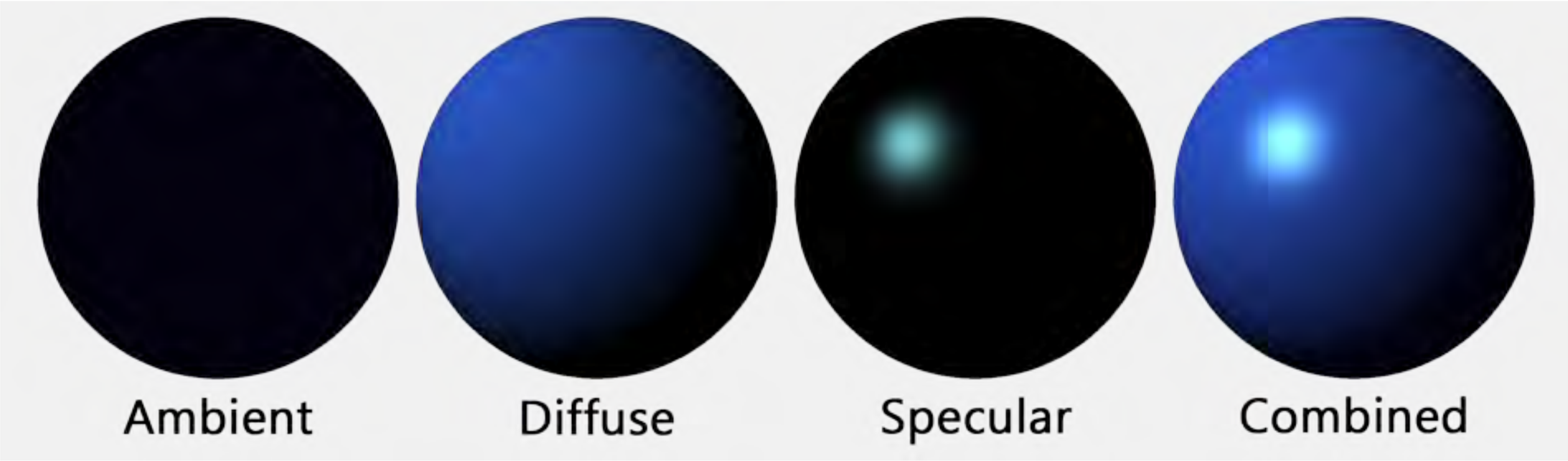


$$cos\theta = \frac{n \cdot l}{\|n\| \cdot \|l\|}$$

Lighting

Ambient, diffuse, specular

$$I = I_{ambient} + I_{diffuse} + I_{specular}$$



[source:clara.io]

Ambient Reflection

K_a Ambient reflection coefficient

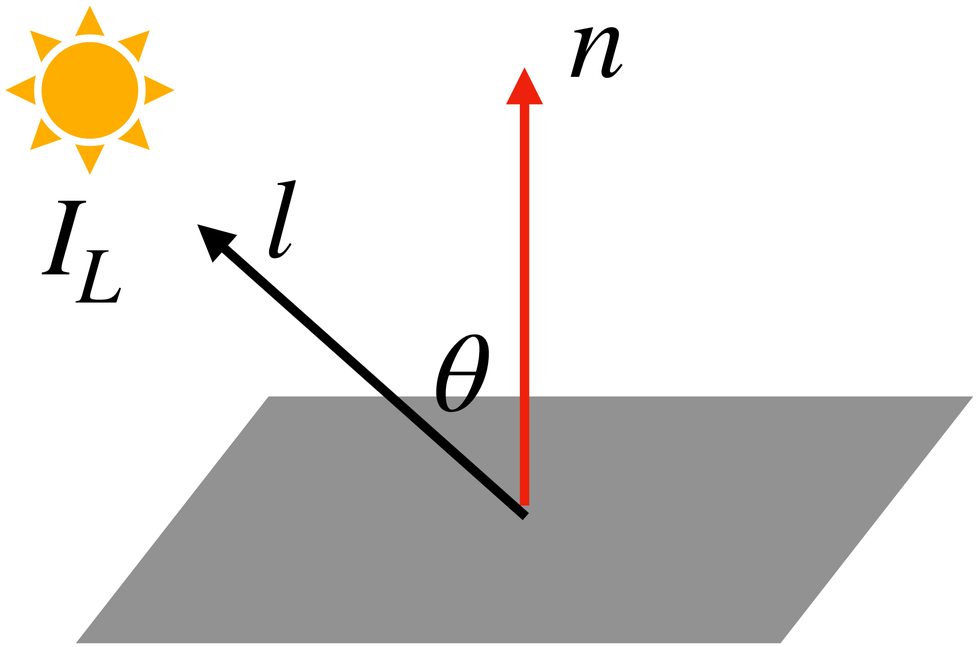
I_L Incoming light intensity

$$I_a = K_a I_L$$

Ambient doesn’t depend on the normal, the light direction, or the view direction

Diffuse reflection

Lambertian model



K_d diffuse reflection coefficient

I_L Incoming light intensity

$$I_d = I_L K_d max(0, cos\theta)$$

Specular Reflection

Reflected light bounces off at the same angles as the incident ray



Diffuse

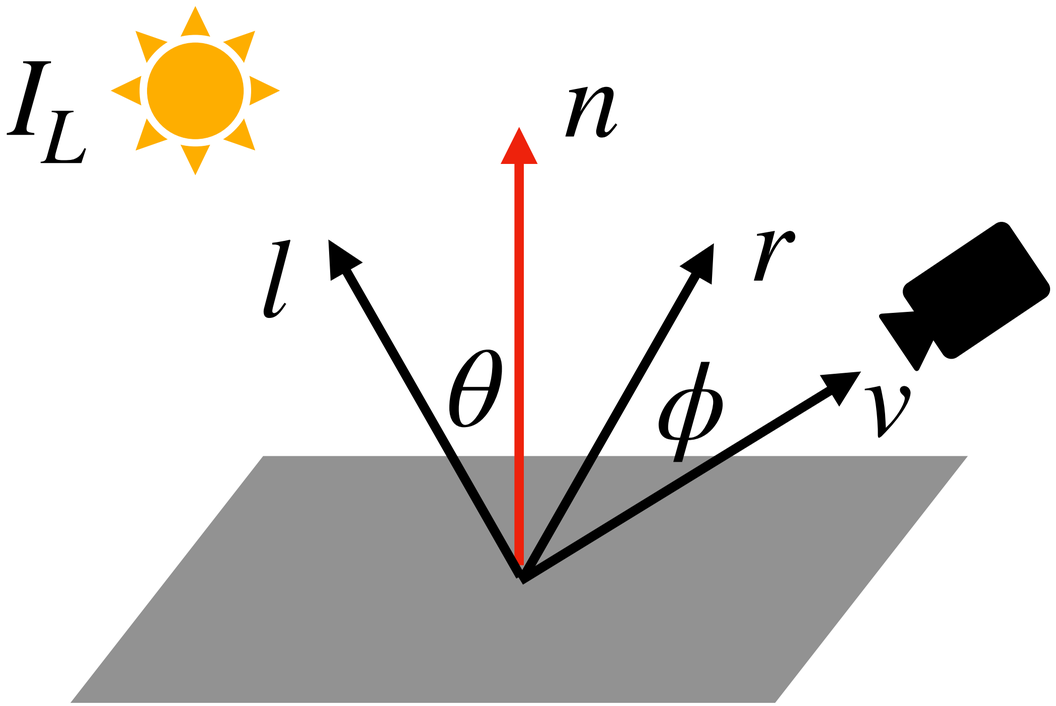


[source:Wikipedia]

Specular

Specular reflection

Phong model



K_s Specular reflection coefficient

I_L Incoming light intensity

α Shininess coefficient(specular exponent)

ϕ Angle between r and v

$$I_s = I_L K_s (max(0, cos\phi))^{\alpha}$$